

CarbonLens User Manual

Browser-based CO₂ Storage Screening & Simulation Studio

Version 3 — POC Release

What Is CarbonLens?

CarbonLens is a fully browser-based CO₂ storage assessment studio. No installation. No backend server. Every calculation — thermodynamics, geomechanics, simulation — runs live in your browser using scientific models calibrated against published field data.

It takes a geologist or engineer from raw formation parameters to a regulator-ready screening report in one session.

Who Is This For?

Government Agency / Regulatory Body

"We need to assess our country's CCS storage potential and evaluate permit applications."

Your workflow:

1. **Overview** → understand the project scope and CO₂ status
2. **Formation** → load a preset formation in your jurisdiction (e.g. Malay Basin, Niger Delta, Nile Delta) or enter your own well data
3. **Geomechanics** → confirm injection pressures stay below MAIP — this is your regulatory safety gate
4. **Simulation** → run storage capacity estimates (DOE P10/P50/P90 range) with trapping analysis
5. **Leakage** → review legacy well cement integrity risk in the Area of Review
6. **Screening** → get a red/amber/green bankability and site readiness score
7. **Jurisdiction** → toggle your regulatory framework (EU ETS, US EPA Class VI, Malaysia PCPP)
8. **Export** → download the executive summary PDF and permit pre-application report

What CarbonLens gives you:

- Independent capacity estimates not supplied by the project developer
 - Geomechanical risk flags before a permit is considered
 - A defensible audit trail (DOE Goodman 2011 methodology, cited references)
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Research / Academic Group

"We are studying CCS feasibility and CO₂ behaviour in saline aquifers."

Your workflow:

1. **Formation** → set formation parameters, or load one of the 16 real-world presets (Sleipner, Johansen, Snøhvit, In Salah, SPE11, etc.)
2. **Properties** → inspect Span-Wagner EOS outputs: CO₂ density, viscosity, solubility, diffusion coefficient, IFT — at your formation's T/P conditions
3. **Geology** → define stratigraphy, facies, fault planes, and caprock geometry for the 3D visualiser
4. **Simulation** → run the plume model; watch residual/solubility/mineral trapping evolve over time
5. **Validation** → compare your results against Sleipner field data (Boait 2012, Furre 2017), or run the physics sanity suite and preset analytical benchmarks
6. **History Matching** → calibrate the model against observed field data (e.g. seismic-derived plume areas)
7. **Methodology** → read the scientific basis for every calculation; all equations are cited

What CarbonLens gives you:

- In-browser ML-augmented property prediction (MARS model, trained on CCS thermodynamic data)
- Physics sensitivity validation: confirm that halving thickness halves P50 capacity within $\pm 15\%$
- Benchmark reproducibility: Sleipner plume radius comparison at years 4, 8, 12
- Export simulation time-series as JSON/Excel for external analysis

Petroleum / CCUS Engineer

"I need to size an injection project, assess geomechanical risk, and produce a screening report for a client."

Your workflow:

1. **Formation** → load the target formation preset (e.g. Gorgon, Alberta Basin, Abu Dhabi) — wells are auto-optimized to the maximum safe injection rate
2. **Geomechanics** → review 6-check geomechanical risk assessment: caprock integrity, safety factor, Mohr-Coulomb failure, MAIP margin; use the optimizer to find the safe injection rate automatically
3. **Simulation** → run the full dynamic simulation with animated plume growth, trapping evolution, DOE capacity range
4. **Economics** → estimate NPV, breakeven carbon price, CapEx/OpEx, and project IRR
5. **Leakage** → quantify legacy well risk in the AoR; add well count and cement condition data
6. **Monte Carlo** → propagate uncertainty across all input parameters to generate probability distributions of capacity and containment
7. **Registry** → log the project to the storage registry with location, operator, and status
8. **Export** → generate the full export package: executive PDF, permit pre-application, JSON data, Excel time-series, 3D viewer screenshot

What CarbonLens gives you:

- Validated pressure model (Theis radial flow with multi-well superposition)
- Auto-optimizer that finds the maximum injection rate consistent with caprock safety
- PwC/Deloitte-grade executive summary PDF (light theme, print-safe, cited methodology)

- Regulatory permit pre-application template — pre-filled from your simulation data

Panel-by-Panel Reference

1. Overview

What it is: Project dashboard showing active wells, status, and a summary of the last simulation result.

Key actions: Create/load a project, set project name, review stored CO₂ and capacity utilisation at a glance.

For agencies: Quick audit snapshot before diving into detail panels.

2. Properties (Fluid Properties)

What it is: Real-time CO₂ and brine property calculator using:

- **Span-Wagner EOS** for CO₂ density and viscosity
- **Duan-Sun model** for CO₂ solubility in brine
- **Garcia (2001)** for brine density
- **MARS ML model** for interfacial tension (IFT)

Key outputs: CO₂ density (kg/m³), CO₂ viscosity (Pa·s), brine density, IFT (mN/m), CO₂ solubility (mol/kg brine), diffusion coefficient

Why it matters: Storage capacity is proportional to CO₂ density. A formation at 100°C/28 MPa stores considerably less CO₂ per cubic metre than one at 37°C/10 MPa. This panel tells you exactly how much.

For researchers: Applicability domain (AD) assessment flags when your T/P/salinity conditions are outside the MARS model's training domain — outputs are marked accordingly.

3. Formation

What it is: Primary input panel for geological parameters and well configuration.

Parameters:

Parameter	What it controls
Depth (m)	Reservoir pressure, temperature gradient, CO ₂ phase
Thickness / Storage Interval (m)	Linear effect on storage capacity ($P50 \propto h$)
Porosity (fraction)	Linear effect on storage capacity ($P50 \propto \phi$)
Permeability (mD)	Injection pressure ($\Delta P \propto 1/k$) — not capacity
Area (km ²)	Linear effect on storage capacity ($P50 \propto A$)

Parameter	What it controls
Net-to-Gross	Used in containment probability; capacity uses gross pore volume per DOE
Pressure / Temperature	CO ₂ density, phase state, Mohr-Coulomb calculations
Salinity	CO ₂ solubility (higher salinity → lower solubility)

Presets: 16 real-world formations. The active preset is highlighted in solid accent colour with a white dot and a pulsing indicator card below. Selecting a preset auto-optimizes your wells.

Wells: Add up to 5 injection wells. Each well has an injection rate, ramp-up/ramp-down schedule, and position in the reservoir.

Important: Thickness and porosity directly scale P50 capacity. Permeability affects injection pressure but NOT capacity. This is a common confusion — the simulator respects the DOE Goodman 2011 framework.

4. Geology

What it is: 3D geological model builder. Define stratigraphy (rock layers), faults, caprock geometry, and facies distribution.

Key actions: Add geological layers with lithology (sandstone, shale, carbonate), set fault planes, toggle caprock geometry. The model feeds the 3D reservoir viewer.

For engineers: Use this to visually confirm the structural trap before running simulation.

5. Geomechanics

What it is: Geomechanical risk assessment for the injection scenario.

6 safety checks:

Check	What it tests	Pass condition
Caprock fracture ratio	Injection P vs caprock fracture P	Ratio < 0.9
Safety factor	Total stress vs effective stress	SF ≥ 1.5
Mohr-Coulomb failure	Shear stress on caprock	Margin > 0 MPa
MAIP margin	How close to Maximum Allowable Injection Pressure	> 20% headroom
Shear reactivation	Fault slip potential	Low risk
Surface heave	Induced uplift	< 5 mm

MAIP definition: $0.9 \times \text{fracture pressure}$ — the regulatory standard (Hubbert-Willis). Sleipner injects at ~50% of fracture pressure.

Auto-optimizer: If checks fail, the "Apply Safe Rate" button recalculates and applies the maximum well rates that satisfy all 6 constraints simultaneously.

Connected to Formation: When you change injection rate in the Formation panel, Geomechanics updates instantly. The Theis pressure model uses your full injection rate (no ramp factor — safety validation always checks peak rate as worst case).

6. Simulation

What it is: Dynamic CO₂ plume simulation with animated results.

Engine: Theis transient radial flow (multi-well superposition) + Span-Wagner EOS + DOE Goodman 2011 capacity framework.

Key outputs:

Output	Definition
Stored CO ₂ (Mt)	Cumulative injection at current year
P50 Capacity (Mt)	DOE statistical estimate: $A \times h \times \phi \times 2.0\% \times \rho_{\text{CO}_2}$
DOE Range	P90 (conservative) / P50 (best) / P10 (optimistic)
Utilisation (%)	Stored / P50 — > 100% signals overpressure risk
Stor. Eff.	Always 2.0% Cc — shown for transparency (DOE Goodman 2011)
CO ₂ Trapping	Residual (pore snap-off), Dissolved (into brine), Mineral (geochemical), Mobile (still migrating)

P10/P90 convention: CarbonLens uses **petroleum engineering convention**: P10 = optimistic (only 10% of outcomes exceed this), P90 = conservative (90% exceed this). This is the opposite of the raw DOE Cc coefficient ordering — the display has been corrected.

Run controls: Play, Pause, Resume, Stop, Re-run, speed multiplier (1x–10x)

Lock Baseline: After a complete run, click "Baseline" to lock the result. Then modify a parameter and re-run — the Validation panel's Δ Params tab will show whether the change produced the physically expected outcome.

7. History Matching

What it is: Calibrates the simulator against observed field data (e.g. CO₂ plume area from seismic surveys).

How it works: Runs the forward model at multiple parameter combinations (permeability, porosity, injection rate) and minimises the mismatch between simulated and observed outputs.

For researchers: Use with Sleipner seismic data (Layer 9 area vs time) to demonstrate model calibration. The matched parameters are more defensible than raw estimates.

8. Economics

What it is: Financial feasibility model for the CCS project.

Key outputs: Total CapEx, OpEx/year, NPV at various carbon prices, breakeven CO₂ price (\$/tonne), project IRR.

For agencies: Use to assess whether a proposed project is financially viable at your jurisdiction's carbon price. A project that requires \$180/tonne to break even at current \$50/tonne prices needs policy support.

For engineers: Quick client-facing sanity check before a full techno-economic study.

9. Leakage

What it is: Legacy well leakage risk assessment in the Area of Review (AoR).

Risk factors: Number of legacy wells, cement condition (0 = failed, 1 = excellent), well age, injection pressure drive (ΔP above reservoir pressure increases leakage drive through old well pathways).

Key rule: Zero wells in the AoR = zero leakage pathway risk (green banner confirms this). The cement factor only activates if legacy wells exist.

For agencies: This is the AoR analysis required by most regulatory frameworks. A high leakage risk score (>400) should trigger additional cement remediation requirements before permitting.

10. Screening

What it is: Bankability and site readiness scorecard.

Output: Red / Amber / Green assessment across capacity, injectivity, seal integrity, economic viability, and regulatory readiness — the same framework CCS investment committees use.

For agencies: Use this to triage multiple site applications — green sites advance, amber sites need conditions, red sites are rejected at screening.

For researchers: Provides a structured framework to report CCS site assessments in publications.

11. Registry

What it is: Project storage registry log.

What it stores: Operator name, location coordinates, formation name, injected volumes, project status (exploratory / active / closed), monitoring data.

Why it matters: Most national CCS regulatory frameworks (EU CCS Directive, US EPA Class VI, Malaysia PCPP) require operators to maintain and report to a national CO₂ storage registry. This panel generates the reporting data in a standard format.

12. Jurisdiction

What it is: Regulatory framework selector.

Options: EU ETS / CCS Directive, US EPA Class VI, Malaysia PCPP/DOE, Australia CCS Act, generic.

Effect: Switching jurisdiction changes the permit template language in the Export panel and adjusts the screening criteria thresholds to match local regulatory standards.

13. Export

What it is: Document generation hub.

Available outputs:

Export type	Description
Executive Summary PDF	Light-theme, print-safe A4 report. PwC/Deloitte-grade layout with CarbonLens logo, cited methodology, DOE capacity range, trapping analysis. Suitable for board presentations and regulatory submissions.
Permit Pre-Application Report	Pre-filled regulatory report using your simulation data. Language adapts to selected jurisdiction. This is NOT a permit — it is a pre-application screening document.
JSON Data Package	Full simulation results in machine-readable format for integration with external tools
Excel Time Series	Year-by-year injection, trapping, and pressure data
3D Screenshot	PNG export of the current 3D reservoir viewer state

14. Methodology

What it is: Scientific reference panel — every equation, model, and coefficient used in CarbonLens with its literature citation.

Why it matters: A simulation tool without cited methodology is not defensible in a regulatory or peer-review context. Every number in CarbonLens traces to a published source.

15. Validation

What it is: Four-tab quality assurance panel.

Tab	What it does
Sanity	Physics checks on current result: supercritical T/P, CO ₂ density range, injection < MAIP, DOE formula consistency
Compare	Automatically compares your result against the active preset's analytical reference baseline. Shows which parameter changes explain the deviation.
Presets	Batch analytical scan of all 16 preset formations — validates CO ₂ phase state, density, expected P50, MAIP
Δ Params	Sensitivity validation: lock a baseline, modify a parameter, re-run — confirms result changes as physics predicts (e.g. halving thickness → P50 drops ~50%)
Sleipner	Direct comparison against Sleipner Utsira field measurements (Boait 2012, Furre 2017)

16. Monte Carlo

What it is: Uncertainty propagation. Runs hundreds of simulations across probability distributions for uncertain inputs (porosity, permeability, thickness) to generate P10/P50/P90 distributions.

For agencies: Replace a single "best estimate" with an honest uncertainty band. A P90 capacity of 50 Mt with P10 of 200 Mt tells a very different story than a point estimate of 100 Mt.

The Permit Section — Dubai Competition Context

Short answer: Yes, keep it — and it is one of your strongest differentiators.

Here is why:

The **Export > Permit Pre-Application Report** is not a regulatory permit — it is a pre-application screening report template pre-filled with your simulation data. Most CCS regulatory frameworks (EU, US, Malaysia, Australia) require operators to submit a site characterisation report *before* a permit application is considered. CarbonLens generates the draft of that document automatically.

For the Dubai competition (ADIPEC / COP28 / similar MENA energy competition context):

1. **Abu Dhabi has a specific ADNOC-2030 CCS target** — the Abu Dhabi Basin preset in CarbonLens directly addresses this. A regulator or developer in the UAE can load the Abu Dhabi Basin preset, run a simulation, and export a pre-application report in one session. That is a demonstration of immediate commercial value.
2. **The MENA CCS market is nascent** — most projects are in the permitting phase, not operational. A tool that accelerates the pre-permitting workflow is directly relevant.

3. **Competition judges are looking for real-world applicability** — showing a complete workflow from formation parameters → simulation → permit document is far more compelling than stopping at capacity numbers.
4. **The jurisdiction toggle** — switching from Malaysia PCPP to UAE and showing how the permit language changes accordingly is a live demo moment that resonates with an international panel.

Recommended demo flow for Dubai:

1. Load **Abu Dhabi Basin** preset → note auto-optimized wells
2. Run simulation → show P50 capacity, DOE range, trapping analysis
3. Switch jurisdiction to UAE/Abu Dhabi
4. Export Executive Summary PDF → show print-ready report with CarbonLens logo
5. Export Permit Pre-Application → show pre-filled regulatory document
6. Open Validation → Compare tab → show result vs analytical reference baseline

This 5-minute demo covers the full value chain: geological assessment → engineering simulation → regulatory output. That is the "promise land."

Common Questions

Q: Why does changing permeability not change storage capacity?

A: Permeability controls how easily CO₂ flows into the rock (injection pressure and rate), but it does not change how much space is available. Capacity is set by pore volume × CO₂ density × efficiency factor. Doubling permeability means you can inject faster at lower pressure — but the total storage potential is the same.

Q: Why is my P50 capacity much smaller than the formation description suggests?

A: The formation descriptions quote the *total* storage potential of the entire geological unit (sometimes thousands of km²). CarbonLens simulates your *project area* (the Area parameter in the Formation panel). Set Area to the full formation area if you want the basin-scale estimate.

Q: What does "Cc = 2.0%" mean in the results?

A: This is the DOE P50 storage efficiency coefficient from Goodman et al. (2011). It means that on average, 2% of the gross pore volume in a saline aquifer is accessible for CO₂ storage at the P50 (most-likely) estimate. P90 (optimistic, 10% probability of exceedance) uses 5.5%, P90 conservative uses 0.51%.

Q: Why is my injection pressure flagged even at low rates?

A: Check the Geomechanics panel — the issue is usually that the formation pressure (depth × 0.0433 MPa/m) is already close to the fracture pressure. The optimizer in Geomechanics will find the maximum safe rate automatically.

Q: Can I use CarbonLens results in a regulatory submission?

A: CarbonLens is a screening and pre-application tool, not a certified regulatory simulator. Results should be used to inform early-stage feasibility and pre-application reports. Site-specific full-physics reservoir simulation (e.g. ECLIPSE, CMG) is required for formal permit applications. The export report includes this disclaimer.

CarbonLens — MSc Research, Universiti Teknologi PETRONAS, Malaysia | Preliminary screening only
Scientific basis: DOE Goodman et al. (2011), Span-Wagner (1996), Duan-Sun (2003), Furre et al. (2017),
Boait et al. (2012)